

LABORATORY REPORT

PHY 110

Engineering Physics



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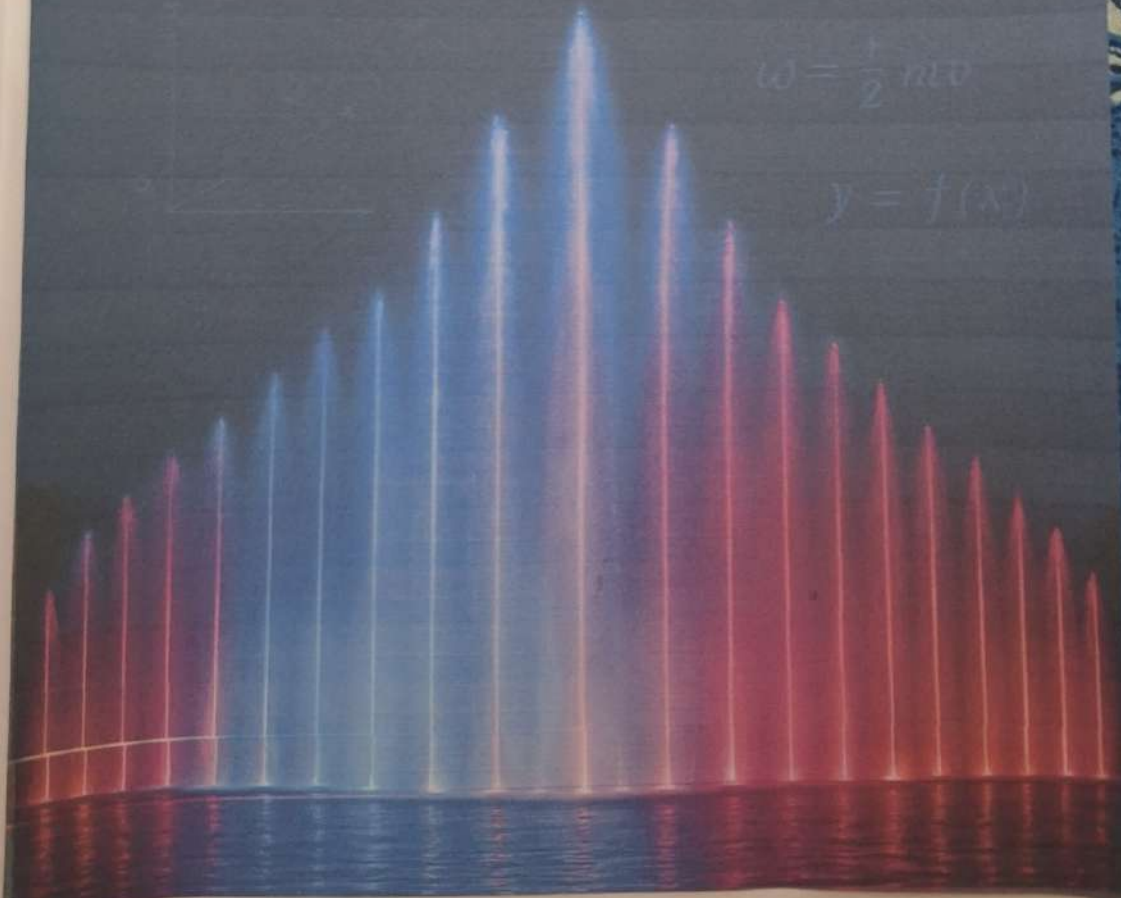
Topic:- Physics Behind Musical Fountains.



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Course:	<u>B.Tech CSE(Core)</u>		
Class/Section:	<u>K25ER</u>		
Roll.No. :	<u>RK25ERB53</u>	Date of Submission:	<u>04/11/2025</u>

Musical Fountains

A Study of Physics Behind Beauty



Aim : To understand and Analyze the physical principles

mainly fluid mechanics, Acoustic (Sound), and optics —
that govern the operation of musical fountain and
theoretically calculate relationships between pressure, velocity
and jet height.

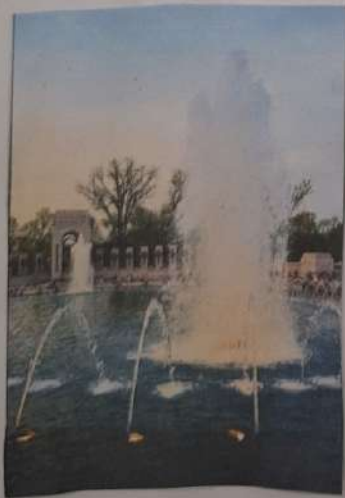
Introduction: A musical fountain is a water display
that synchronizes the motion of water jets and
colored lights with music. It operates through the
integration of fluid mechanics, Acoustic (Sound), optics,
and control systems. Each of these branches of physics
contributes to the beautiful coordination seen in
musical fountains. In this project, we study the
physics behind how water jets rise and fall with
changing musical intensity.

Theory:

1. Fluid Mechanics
2. Acoustics (Sound)
3. Optics.

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Other looks of Musical fountains : —





Source: Chat GPT.

Theory:

1. Fluid Mechanics:

A. Description: Fluid Mechanics is the branch of physics that studies fluids (liquids, gases, and plasmas) and the forces acting on them, covering both fluids at rest (fluid static) and in motion (fluid dynamics). This field applies the laws of force and motion to understand fluid behavior in engineering, such as

and has wide-ranging applications in engineering, designing airplanes, dams, and pipe systems, as well as in fields like meteorology and oceanography. Key concepts include viscosity, surface tension, buoyancy and principles like Bernoulli's Principle.

Key Aspects of Fluid Mechanics :

(a) Fluid : In physics, a fluid is a substance that flows and continuously deforms under an applied shear stress, lacking a fixed shape. This term includes liquids, gases, and plasmas because

their particles can move freely and change position.

(b) Fluid states : Fluid states refer to the states of matter where a substance can flow and take the shape of its container, which includes liquids and gases.

Liquid : A fluid state with a fixed volume but no fixed shape. Particles are held together by intermolecular forces, allowing them to move around but not escape completely. Ex: water, oil, blood.

Gases : A fluid state with no fixed volume or shape. Particles are far apart and move randomly, expanding to fill the entire container they are in. Ex: Air, Oxygen and Steam.

C) Fluid Dynamics : The branch of physics that studies the movement of liquids and gases, and it has applications in fields like weather forecasting, engineering, and biology. It involves principles like viscosity, Bernoulli's theorem and the continuity equation like velocity, pressure, fluid motion.

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(d) Fluid Kinematics : Fluid kinematics is the study of fluid motion without considering the forces that cause it. It describes how fluids move in terms of velocity, acceleration, and flow patterns. The motion can be steady or unsteady, uniform or non-uniform. It uses concepts like streamlines, pathlines, and streaklines to visualize flow. Mathematical tools such as the continuity equation, stream function and velocity potential are used to analyze fluid motion.

B. Application of Fluid Mechanics :

a) Control of Water Flow : Determines the speed and

height of water jets using fluid pressure and flow rate

- b) Nozzle design: Shapes the water streams into desired patterns and effect.
- c) Pump systems: Maintain continuous water circulation and pressure.

d) Synchronization: uses Computer-controlled fluid dynamics to match water motion with music and lights.

e) Energy efficiency: Optimizes water and power usage for smooth and artistic operation.

C. Key Concepts:

a) Viscosity: Viscosity is the property of a fluid that resists the relative motion between its layers. It is a measure of a fluid's internal friction or thickness.

Role of Viscosity in Musical fountains:

→ Smooth flow: Controls how easily water flows through pipes and nozzles.

→ Jet shape: Affects the height, thickness, and

stability of the water.

- Energy efficiency: Low Viscosity fluids (CH_2O) reduce friction losses in pumps and pipes.
- Spray patterns: Helps in forming clear, consistent

water shapes for visual effects.

- System performance: Ensures proper coordination b/w water flow, lights, and music.

b) Surface Tension: It is the property of a liquid that causes its surface to behave like a stretched elastic sheet. It occurs due to cohesive forces b/w liquid molecules at the surface.

Role of Surface Tension in Musical fountain:

- Helps maintain smooth and stable water jets
- Affects the formation of water droplets and sprays.
- Ensures clean separation of water streams from nozzles.

→ Contributes to the visual clarity of patterns and shapes in the fountain

and splashing of misting during

→ Reduces maintenance
Operation.

c) Buoyancy : It is the upward force exerted by a fluid on an object submerged in it, opposing the weight of the object. This happens because of the difference in pressure at different depths in the fluid.

Role of Buoyancy in Musical Fountain :

→ Supports floating platforms : Buoyancy keeps pumps, nozzles, and lighting systems afloat in water-based musical fountains.

→ Maintains stability : Proper buoyant balance prevents platforms from tilting, ensuring water jets shoot straight.

→ Affect Equipment placement : Submersible components like lights or sensors use neutral buoyancy to stay in position underwater.

→ Enables floating decorations : Buoyant decorative elements (like balls or figures) move gracefully with the

water's rhythm.

→ Guides Design Calculations: Engineering use Archimedes' Principle to calculate buoyant force and ensure safe load support.

→ Improve Performance: Water, buoyant setups reduce vibration and enhance synchronization of water, light, and music effects.

d) Bernoulli's Theorem: It states that for an incompressible, non-viscous fluid in steady flow, the sum of pressure energy, kinetic energy, and potential energy per unit volume remains constant along any streamline. Mathematically:

$$P + \frac{1}{2}\rho v^2 + \rho gh = \text{Const.}$$

where →

- P = Pressure energy per unit volume.
- ρ = density of the fluid.
- v = Velocity of the fluid.

- g = acceleration due to gravity
- h = height above a reference level.

Explanation

- If the speed of fluid increases, its pressure decreases, and vice versa.
- This is because the total energy ($P_{\text{press}} + \text{kinetic} + \text{potential}$) stays constant along a streamline.

Role of Bernoulli in Musical fountain :

- Helps water shoot up : Bernoulli's principle explains how water moves faster when pressure from the pump increases.
- Decides jet height : More pressure makes the water go higher ; less pressure makes it lower.
- keeps flow smooth : Nozzles are shaped using this rule so the water comes out in neat, clean streams.

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the jets rise and fall with the rhythm of the music

→ Saves Energy: It helps use the right amount of pump power. So energy isn't wasted.

→ Creates different patterns: The principle is used to make different water shapes like mist, fans, or spirals.

Derivation:

(i) Bernoulli eqn with pump head and losses.

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + Z_1 + H_{\text{pump}} - h_L = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + Z_2$$

(ii) Use $P_2 = P_{\text{atm}}$ as reference (Set Pressure at exit = 0) and simplify usually V_1 is much smaller than V_2 so

$\frac{V_1^2}{2g} \approx 0$ and often $Z_1 \approx Z_2$ so elevation terms

Cancel —

This gives —

$$\frac{V_2^2}{2g} = H_{\text{pump}} - h_L$$

iii) Solve for exit velocity V_2 .

$$V_2 = \sqrt{2g(H_{\text{pump}} - h_L)}$$

iv) Relationship between velocity and jet height (Energy Conservation)
Kinetic energy of the jet converts into potential energy at maximum rise; ideal max height h_{max} (above nozzle)

is:

$$h_{\text{max}} = \frac{V_2^2}{2g}$$

$$\Rightarrow h_{\text{max}} = H_{\text{pump}} - h_L$$

Conclusion: Thus, by Applying Bernoulli's theorem, we find that the height of water jets in musical fountain depends on the pump pressure (head) and Energy losses in the system.
Higher pump head and smoother flow (less loss) produce taller and smoother fountain jets

B. Acoustic (Sound wave):

1. Introduction: In simple terms, anything described as acoustic deals with the production, transmission, or effects of sound waves. Music consists of sound waves. Loudness (Amplitude) Controls Pump Pressure; thus, jet height

H2 A (Completed)

2. Acoustic Function in the System :

- Sound Detection : Acoustic sensors or microphones detect the sound or music being played.
- Signal Conversion : The sound waves are converted into electrical signals.
- Control Action : These signals are sent to a controller, which adjusts the speed of pumps or opening of valves.
- Jet movement : The water jets rise and fall according to the intensity and rhythm of the sound.

→ Visual Effect : This creates a synchronized display where water dances to the beats of the music.

In short :

Acoustic help the musical fountains convert sound energy into motion, making the water move in time with the music.

3. Acoustic Aspects of Musical Fountains :

microphones or

- Sound Sensitivity: The fountain uses acoustic sensors to pickup music or sound signals.
- Frequency Response: Different frequencies (bass, mid, treble) control diffⁿ water jets or light effects.
- Amplitude Control: The loudness of sound (audio) decides how high the water jets rise.
- Signal Processing: The sound signals are processed by controller or computer to match water movement with rhythm.

→ Synchronization: All jets and lights move in harmony with the music; creating a pleasing visual and acoustic experience.

In short:

The acoustic aspects help the fountain sense, analyze, and respond to sound; making the water and light dance beautifully to music.

4. Conclusion: Acoustic play a vital role in musical fountains by linking sound with motion and light. The music rhythm, frequency, and amplitude are analysed and linked to different effects. Different

to control water jets and lighting effects. Sounds trigger specific water patterns, creating harmony between audio and visuals. This synchronization turns music into a dynamic, moving art form. Overall, acoustic make musical fountains both technically advanced and artistically captivating.

C. Optics :

1. Introduction : Optics is a branch of physics that deals with the study of light, its properties and behaviours when it interacts with different materials. It explains how light is reflected, refracted, absorbed or scattered to produce various visual effects. Optics is widely used in technology such as lenses, lasers and lighting systems. In applying like musical fountains, optics helps create beautiful colourful lighting and visual patterns that synchronize with music and water movement. Thus, optics connects science and art to enhance human visual experience.

2. Role of Optics in Musical Fountains.

- Optics enhances the visual beauty of musical fountains through the use of light and colour.
- LED lights, lenses, and colour filters create different lighting effects and patterns.

→ The reflection and refraction of light on water jets produce sparkling and colourful visuals.

→ Light effects are synchronized with the rhythm and mood of the music.

→ Optical systems help highlight water movements and make them appear more dynamic.

→ Overall, optics combines science and art to turn fountains into spectacular visual displays.

3. Key Aspects of Optics in Musical Fountain:

(i) Light Source : The main light sources in musical fountains are LED lights, lenses, and sometimes projectors. These lights create colourful effects that enhance the beauty of the water display. They sync up with the music and water movement.

are synchronized with
to produce dynamic visual patterns. LED's are
preferred because they are bright, energy-efficient and
long-lasting.

ii) Color Control: Color control in musical fountains is achieved using RGB LED light and advanced control system. Different colours are programmed to match the mood, rhythm, and tone of the music. By mixing red, green, and blue light, a wide range of colors can be created. This synchronization of color with sound and water movement makes the fountains.

iii) Synchronization: Lighting patterns are timed precisely with music beats and water movements.

iv) Reflection and Refraction: Reflection and refraction play a key role in creating the sparkling beauty of musical fountains. When light hits the surface of water, some of it is reflected, making the jets shine and shimmer. The rest of light refracts, bending as it passes through water droplets to form glowing, colorful effects. These optical phenomena make the fountains dynamic and full of motion.

v) Angle and intensity : The angle and intensity of light in musical fountains determine how water jets are illuminated and highlighted. By adjusting the angle, lights can create different reflections, shadows and focus points on the moving water. Intensity control changes the brightness to match the mood or rhythm of the music. Together, they enhance the depth, color, and drama of the fountain's visual display.

vi) Technology Integration : Technology is the backbone of musical fountains, allowing music, water, and light to move in perfect coordination. The system begins with digital signal processing (DSP), which analyzes the sound's rhythm, frequency, and intensity. That control's pumps, valves, and LED lights. A computerized control system or programmable logic controller (PLC) manages all these elements in real time. It ensures that each water jet, color

change, and light pulse matches the beat and mood of the music. Software program are used to design and program different fountain "shows" allowing modern fountains also include wireless IoT or sensor-based systems, enabling remote operation and automatic synchronization.

Through this integration of electronics, software and hydraulics, musical fountains become an advanced display of art and engineering, creating mesmerizing experience that blend technology with creativity.

vii) Aesthetic Appeal: Optical effects make the fountain visually attractive and create a magical, immersive experience.

Conclusion: Optics in musical fountains deals with the use of light and color to enhance water displays. LED's, lasers and projections create colorful effects that move in sync with the music.

Light undergoes reflection and refraction in water, producing glowing and sparkling patterns. By adjusting angle, color, and intensity, designers match visuals with rhythm and mood. Thus, optics makes musical fountains vibrant, artistic and visually mesmerizing.

D. Result of musical fountains is a beautiful Synchronization of water, light, and music that creates a visually and emotionally Captivating show.

They enhance the beauty of Public Spaces and attract tourists. Musical fountains Provide relaxation and entertainment to viewers.

They also showcase a blend of art, technology, and Creativity.

